

Emil Khabiboulline

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Researcher in quantum computing theory with expertise in communication, sensing, security, simulation, and algorithms, and familiarity with quantum optics and many-body physics relevant to experiment.

EMPLOYMENT

National Institute of Standards and Technology & University of Maryland College Park (Joint Center for Quantum Information and Computer Science)	September 2023 - Present
<i>National Resource Council Postdoctoral Associate</i>	<i>College Park, MD</i>
Harvard University, Department of Physics	February - May 2023
<i>Postdoctoral Fellow</i>	<i>Cambridge, MA</i>
Amazon Quantum Computing	June - September 2022
<i>Research Scientist Intern</i>	<i>Pasadena, CA</i>
Kavli Institute for the Physics and Mathematics of the Universe	June - August 2013
<i>Visiting Researcher</i>	<i>Kashiwa, Japan</i>
Fermi National Accelerator Laboratory	June - August 2011, June - July 2012
<i>Seasonal Employee</i>	<i>Batavia, IL</i>

EDUCATION

Harvard University	August 2016 - January 2023
<i>Ph.D. and M.A. in Physics</i>	<i>Cambridge, MA</i>
· <i>Dissertation:</i> Quantum Communication and Thermalization, From Theory to Practice. · <i>Committee:</i> Mikhail Lukin (chair), Aram Harrow, Daniel Jafferis, Ashvin Vishwanath. · <i>Funding:</i> National Science Foundation Graduate Research Fellow.	
California Institute of Technology	September 2012 - June 2016
<i>B.S. in Physics</i>	<i>Pasadena, CA</i>
· <i>Funding:</i> Stamps Scholar, Dunham Scholar, Fermi Research Alliance Scholar. · <i>Honors:</i> Frank Teruggi Memorial Award, Lucy Guernsey Service Award, SanPietro Travel Prize.	
Illinois Mathematics and Science Academy	August 2009 - June 2012
<i>High School Diploma</i>	<i>Aurora, IL</i>
· <i>Honors:</i> National Merit Finalist, National AP Scholar.	

PATENTS

- E. T. Khabiboulline, J. S. Sandhu, M. U. Gambetta, M. D. Lukin, and J. Borregaard. Efficient quantum voting with information-theoretic security, U.S. Ser. No. 18/069,179.

PUBLICATIONS

- Daniel Alabi and Emil T. Khabiboulline. Quantum Advantage for Two-Party Differential Privacy, ePrint:2026/1669.
- Yousef K. Chahine, Chaohan Cui, William DeRocco, Daniel Gottesman, Saikat Guha, Emil T. Khabiboulline, Zhenning Liu, Brittany McClinton, Jayadev Rajagopal, Fredrik Rantakyro, J. Gabriel Richardson, Stephen Ridgway, Aqil Sajjad, and Joohyung Song. Eliminating photon transport in long-baseline optical interferometry using quantum memories, arXiv:2608.10078.
- D. J. Spencer, K. Shi, E. T. Khabiboulline, G. Alagic, and A. V. Gorshkov. Differentially private quantum sensor networks, arXiv:2607.06521.
- Z. Liu, W. DeRocco, S. Gu, E. T. Khabiboulline, S. Choi, A. M. Childs, A. Hook, A. V. Gorshkov, and D. Gottesman. Measuring gravitational lensing time delays with quantum information processing, arXiv:2510.07898. Accepted to TQC 2026.
- A. M. Dalzell, S. McArdle, M. Berta, P. Bienias, C.-F. Chen, A. Gilyén, C. T. Hann, M. J. Kastoryano, E. T. Khabiboulline, A. Kubica, G. Salton, S. Wang, and F. G. S. L. Brandão. *Quantum Algorithms: A Survey of Applications and End-to-end Complexities*. Cambridge University Press, Cambridge, April 2025.
- T. Schuster, B. Kobrin, P. Gao, I. Cong, E. T. Khabiboulline, N. M. Linke, M. D. Lukin, C. Monroe, B. Yoshida, and N. Y. Yao. Many-Body Quantum Teleportation via Operator Spreading in the Traversable Wormhole Protocol. *Phys. Rev. X*, 12:031013, July 2022.
- E. T. Khabiboulline, J. S. Sandhu, M. U. Gambetta, M. D. Lukin, and J. Borregaard. Efficient Quantum Voting with Information-Theoretic Security, arXiv:2112.14242.
- E. T. Khabiboulline, J. Borregaard, K. De Greve, and M. D. Lukin. Quantum-assisted telescope arrays. *Phys. Rev. A*, 100:022316, August 2019.
- E. T. Khabiboulline, J. Borregaard, K. De Greve, and M. D. Lukin. Optical Interferometry with Quantum Networks. *Phys. Rev. Lett.*, 123:070504, August 2019.
- S. Peng, R. Zhang, V. H. Chen, E. T. Khabiboulline, P. Braun, and H. A. Atwater. Three-Dimensional Single Gyroid Photonic Crystals with a Mid-Infrared Bandgap. *ACS Photonics*, 3(6):1131–1137, 2016.
- E. T. Khabiboulline, C. L. Steinhardt, J. D. Silverman, S. L. Ellison, J. T. Mendel, and D. R. Patton. Changing Ionization Conditions in SDSS Galaxies with Active Galactic Nuclei as a Function of Environment from Pairs to Clusters. *The Astrophysical Journal*, 795(1):62, 2014.
- E. J. DiMarco, E. Khabiboulline, D. F. Orris, M. A. Tartaglia, and I. Terechkine. Superconducting Solenoid Lens for a High Energy Part of a Proton Linac Front End. *IEEE Transactions on Applied Superconductivity*, 23(3):4100905, June 2013.

PRESENTATIONS

- *Talk*: Seminar at QuSoft, University of Amsterdam and CWI (July 21, 2026).
- *Talk*: Optica ImageSense Congress 2026.
- *Talk (Invited)*: QuAIL Virtual Seminar at NASA Ames Research Center (March 12, 2026).
- *Poster*: Quantum Information Processing (QIP) 2026.
- *Talk (Invited)*: Seminar at Fermilab SQMS Center (January 8, 2026).
- *Talk*: Journal Club of Quantum Information Group at NIST Boulder (July 31, 2025).
- *Talk*: Quantum Networking Seminar at NIST Boulder (July 29, 2025).
- *Poster*: Robust Quantum Simulation Workshop 2025.
- *Poster*: 32nd Annual NIST Sigma Xi Early-Career Poster Presentation (2025).

- *Talk (by coauthor)*: American Physical Society Global Physics Summit 2025.
- *Poster*: Quantum Information Processing (QIP) 2025.
- *Talk (Invited)*: NIST Quantum Networking Seminar (December 17, 2024).
- *Poster*: Advancing Quantum Computation Beyond Gate-Model 2024.
- *Poster*: Theory of Quantum Computation, Communication and Cryptography (TQC) 2024.
- *Talk*: Friday Quantum Seminar at the University of Maryland College Park (October 27, 2023).
- *Talk (Invited)*: JQI-QuICS Special Seminar at Joint Center for Quantum Information and Computer Science (July 26, 2023).
- *Talk (Invited)*: Liang Jiang’s Group Meeting at University of Chicago (virtual) (March 31, 2023).
- *Talk*: Dissertation Defense at Department of Physics, Harvard University (December 2, 2022).
- *Poster*: Quantum Information Processing (QIP) 2021.
- *Talk (Invited)*: Dinner Seminar at Cosmic Dawn Center (July 21, 2020).
- *Talk (Invited)*: Provocateur Presentation at Quantum Leap Challenge: Quantum Sensing Ideas Lab (May 18, 2020).
- *Talk (Invited)*: Quantum Computing Seminar (January 13, 2020) at Brookhaven National Laboratory.
- *Talk*: American Physical Society March Meeting 2019 – Focus Session on Distributed Quantum Computation, Networking, and Information Security.
- *Poster*: Quantum Information Processing (QIP) 2019.
- *Talk (Invited)*: Lunch Seminar (November 8, 2018) at Institute for Theoretical Atomic, Molecular, and Optical Physics (Harvard).
- *Poster*: Quantum Science Gordon Research Conference 2018.
- *Talk*: John Preskill’s Group Meeting at Caltech (May 23, 2018).
- *Talk*: Workshop on Quantum Information (April 23-24, 2018) at Center of Mathematical Sciences and Applications (Harvard).
- *Talk*: American Physical Society March Meeting 2016.
- *Poster*: International Conference on Quantum, Atomic, Molecular, and Plasma Physics 2015.
- *Poster (Chambliss Astronomy Achievement Student Award)*: 223rd Meeting of the American Astronomical Society.
- *Poster (Special Award from CERN)*: Intel International Science and Engineering Fair 2011.

SERVICE

- *Reviewer*: Nature Communications.
- *Referee*: Theory of Quantum Computation, Communication and Cryptography (TQC) 2026.
- *Reviewer*: Phys. Rev. Letters.
- *Reviewer*: ACM Transactions on Quantum Computing.
- *Reviewer*: Phys. Rev. A.
- *Reviewer*: EPJ Quantum Technology.
- *Reviewer*: Proceedings of the National Academy of Sciences.
- *Reviewer*: npj Quantum Information.
- *Referee*: Young Quantum Information Scientists (YQIS) 2024.
- *Referee*: Quantum Information Processing (QIP) 2023.
- *Referee*: QCrypt (Quantum Cryptography) 2022.
- *Reviewer*: IEEE Transactions on Computers.
- *Referee*: Quantum Computing Theory in Practice (QCTiP) 2020.

INSTRUCTION

- *Teaching Fellow*: Harvard Computer Science 127/227 (Cryptography), Fall 2021.
- *Teaching Fellow*: Harvard Physics 271 (Topics in the Physics of Quantum Information), Fall 2020.

CODING

- *Languages*: C++, Python, Julia.
- *Software*: Mathematica, MATLAB, COMSOL Multiphysics.
- *Hardware*: High-performance computing clusters.